

NSCI0032/25-26, Assignment 2:

Molecular Dynamics for Novel SSE Batteries

18th November 2025

1 Introduction

In this assignment you will be requested to read through an article and understand the molecular dynamics simulations and background of the system. Then answer the following.

The paper is found together with the LAMMPS code files on the NSCI0032 repository which you need as usual to pull (`git pull`) in order to update it.

1.1 Getting Started

From the NSCI0032 teaching GitHub repository, import the project folder for the molecular dynamics assignment. The folder will contain:

- A relevant literature paper by Dai et al..¹
- Two LAMMPS input files.

Before attempting the quiz, make sure you are familiar with the contents of these files.

1.2 Report and General Instructions for Submission

You need to submit to Moodle as well as GitHub a document containing two parts, the first is for Part 1 (include the questions and your answers) and Part 2 is a brief report, between 1000 and up to 2000 words - Figures, Tables and captions are not included in the word count - on your results. Your report should include (not necessarily as separate sections):

- an Introduction,
- a brief theoretical Background,
- Methods,
- Results and Discussion,
- Conclusions,
- a Bibliography if needed.

2 Part 1: Theory and Background Quiz [10 marks]

These are to be included in the final report! See Section 3 and 4. The questions do not count for the word count limit!

Each question is weighted at 0.5 a point! and there are 20 questions in total.

2.1 Questions on Theory and Literature

Please read the paper, you do not need to read all of it, but can scan it for the relevant parts, but of course it may help to read it quickly.

1. What are the key advantages of solid state electrolyte (SSE) and what are the main challenges? (Name at least one of each)
2. What is the low temperature stable phase of LLZO?
3. What is the high temperature phase?
4. Compare both and discuss main differences with respect to the diffusion of Li , i.e. with respect to battery applications.
5. Can you discuss the role of Molecular Dynamics in this study? What impact can it bring?
6. Describe the interatomic potential models considered in the paper by Dai et al.,¹ discuss why they were chosen, what other Interatomic potential models could you propose that in your opinion would have been also suitable or even better. (See for example [List INP Database](#) and [Interatomic Potentials](#))
7. What is the range of temperatures considered in this study? What is the phase transition temperature? Are these relevant for the operation of the battery?
8. Which software tool and methods were used by the authors? Compare it to LAMMPS. Propose a study of your own!

2.2 Questions on the LAMMPS model

In the below, refer and mention the appropriate lines of code in the input file.

1. Describe the supercells, i.e. the lattice models (including boundary conditions) used in this work and why is supercell size an important consideration?
2. How many atoms are in a $1 \times 1 \times 1$ and $2 \times 2 \times 2$ LLZO garnet supercell?
3. What is an appropriate time frame for these computations to run? i.e., How many time steps? Why? Please provide evidence in your report.
4. How do we set the start and end temperatures for the simulation? refer to the input file and explain!
5. What is an appropriate dampening T_{damp} for a Noose-Hoover thermostat in LAMMPS?
6. Which two lattice directions are coupled together for this computation? show the relevant command and explain it. Why is this coupled?
7. What simulation data are we interested in printing out for this simulation? What further postprocessing do we need to do?
8. What scale factor must we divide by for a $2 \times 2 \times 2$ supercell to extract the lattice parameters?
9. Which potential model is used? Compare to the paper.
10. What unit style is used and how does this unit style effect the timestep selection?
11. What boundary conditions are used for the simulation. Is it reasonable for solid LLZO and why?
12. For *pair_style buck/coul/long 10.0*, what is the physical meaning of cutoff 10.0?

3 Part 2: Molecular Dynamics Simulation [20 Marks]

After studying the paper, and getting familiar with the input files provided, you are required to partially reproduce Figure 2. from Dai et al. using the lammps input files provided to you as follows:

Choose two temperatures, one from the tetragonal and one from the cubic phase, and calculate the average lattice parameters in each case you need to provide a thorough account of your simulations and calculations in the report. This can be similar to that in the paper, mention the starting temperature, the equilibration, pre and post processing and of course the final lattice parameters, as well as any changes made to the input file. How did you calculate the lattice parameters?

Can you estimate error bars? How?

Hint: you can replace the Temperature in the input file and run manually in a terminal, or directly or use a Python script in a Jupyter notebook.

Include in your report a visualisation of the initial system and the final ones. Explain in your report how you did the visualisation.

You can add simulation animations, additional code etc. to your GitHub repository and describe this in the report.

3.1 Code for Visualization

You can choose one of two methods (or both) as below, but you can use any system you choose, e.g. AtomEye (which is installed under /apps/AtomEye).

3.1.1 Use Jupyter notebook, ASE and ngview

```
1 from ase.io import read
2 from ase.visualize import view
3 traj = read("dump\_all.lammpstrj", index=":")
4 view (traj, viewer = "ngl")
```

Listing 1: visualisation in Jupyter

3.1.2 Use OVITO

We have also installed OVITO which can visualize the data. For this to work, you need to update your .bashrc file as follows:

```
1 #In a terminal shell
2 #Open a terminal either in Jupyter or the Desktop
3 > cd <enter> to go home
4 > cp /public/adham/.bashrc .
5 # close the terminal and open a new one (in Desktop of course) and start ovito bu typing
6 > ovito
```

Listing 2: update your .bashrc

4 Submission

All submissions should be on Moodle, however, you can also add your code to your GitHub repository on the NSCI0032 organisation, make sure to make a new folder for Assignment 2! Here you can push your source code and add a link to your repository and a description in the appendix!

5 Bonus Questions and Tasks [+5 marks]

For bonus marks, consider the following as part of your report:

1. Run the LAMMPS script at several more temperature values in order to reproduce the full phase transition curve. Compare your curve to the ones in literature.

2. How does the choice of supercell size affect the simulation results?
Hint: you may wish to look at the paper by Klenk et al. (2016)² for some ideas.
3. What are the effects of introducing impurities (e.g. Ta) on the lattice parameter curves and garnet properties?
Hint: you may wish to look at the papers by Dai et al. (2022)³ and Wang et al. (2014).⁴

References

- [1] Jin Dai, Qian Chen, Tobias Glossmann, and Wei Lai. "Comparison of interatomic potential models on the molecular dynamics simulation of fast-ion conductors: A case study of a Li garnet oxide $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ". en. In: *Computational Materials Science* 162 (May 2019), pp. 333–339. ISSN: 09270256. DOI: 10.1016/j.commatsci.2019.02.044. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0927025619301193> (visited on 11/20/2025).
- [2] Matthew J. Klenk and Wei Lai. "Finite-size effects on the molecular dynamics simulation of fast-ion conductors: A case study of lithium garnet oxide $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ". en. In: *Solid State Ionics* 289 (June 2016), pp. 143–149. ISSN: 01672738. DOI: 10.1016/j.ssi.2016.03.002. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0167273815301831> (visited on 05/31/2025).
- [3] Jin Dai, Yue Jiang, and Wei Lai. "Study of diffusion and conduction in lithium garnet oxides $\text{Li}_x\text{La}_3\text{Zr}_{x/5}\text{Ta}_{7-x}\text{O}_{12}$ by machine learning interatomic potentials". en. In: *Physical Chemistry Chemical Physics* 24.24 (2022), pp. 15025–15033. ISSN: 1463-9076, 1463-9084. DOI: 10.1039/D2CP00591C. URL: <https://xlink.rsc.org/?DOI=D2CP00591C> (visited on 05/31/2025).
- [4] Yuxing Wang, Ashfia Huq, and Wei Lai. "Insight into lithium distribution in lithium-stuffed garnet oxides through neutron diffraction and atomistic simulation: $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ ($x = 0-2$) series". en. In: *Solid State Ionics* 255 (Feb. 2014), pp. 39–49. ISSN: 01672738. DOI: 10.1016/j.ssi.2013.11.017. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0167273813005961> (visited on 06/03/2025).